These conditions are sufficient to determine the form of the vectors |*i*> and |*o*>, apart from the phase ambiguity. Here is the result:

$$\begin{aligned} |i\rangle &= \frac{1}{\sqrt{2}}|u\rangle + \frac{i}{\sqrt{2}}|d\rangle \\ |o\rangle &= \frac{1}{\sqrt{2}}|u\rangle - \frac{i}{\sqrt{2}}|d\rangle. \end{aligned} \tag{2.10}$$

Exercise 2.2: Prove that |*i*> and |*o*> satisfy all of the conditions in Eqs. 2.7, 2.8, and 2.9. Are they unique in that respect?

It's interesting that two of the components in Eqs. 2.10 are imaginary. Of course, we've said all along that the space of states is a complex vector space, but until now we have not had to use complex numbers in our calculations. Are the complex numbers in Eqs. 2.10 a convenience or a necessity? Given our framework for spin states, there is no way around them. It's somewhat tedious to demonstrate this, but the steps are straightforward. The following exercise gives you a road map. The need for complex numbers is a general feature of quantum mechanics, and we'll see more examples as we go.

Exercise 2.3: For the moment, forget that Eqs. 2.10 give us working definitions for |*i*> and |*o*> in terms of |*u*> and |*d*>, and assume that the components α, β, γ, and δ are unknown:

$$|i\rangle = \alpha|u\rangle + \beta|d\rangle$$

$$|o\rangle = \gamma|u\rangle + \delta|d\rangle.$$